

Nandini Raut

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EDUCATION

- **California State University, Chico**
Bachelor's in Computer Science

August 2021 – May 2024
Chico, California

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Meta Certified UI/UX Design Principles
- IBM Enterprise Design Thinking Practitioner

January 2023
June 2024
March 2020

SKILLS

- **Programming Languages:** Python, C/C++, JavaScript, Rust, Go, Java, Ruby, Swift | **Operating Systems:** Linux (proficient with CLI), Windows, MacOS
- **Site Reliability & DevOps:** Docker, Kubernetes, CI/CD (Jenkins, GitLab), container orchestration, system monitoring (Splunk, Nagios), networking (SSH, DNS, TCP/IP), performance tuning, fault-tolerant design | **Frameworks and Tools:** Flask, Django, Git/GitHub, Prometheus, Grafana, Helm
- **Web Technologies:** AngularJS, React.js, Node.js, HTML5, CSS3, Bootstrap, WordPress, Express.js | **Database:** SQL, MongoDB, Firebase
- **Cloud Platforms:** AWS (Amazon Web Services), Microsoft Azure, Google Cloud, IBM Cloud | **Version Control:** Git/GitHub
- **Soft Skills:** Problem-solving, Collaboration, Communication, Time management

EXPERIENCE

- **Systems Engineer**
UPS

September 2024 – Present
Atlanta, Georgia

- Developed scalable server-side scripts using Python and Bash to automate infrastructure tasks, reducing manual effort and increasing operational efficiency. Conducted root cause analysis and resolved complex infrastructure issues, applying strong knowledge of object-oriented programming and system debugging.
- Monitored system health and performance using tools like Splunk and Nagios, proactively identifying bottlenecks and minimizing downtime for mission-critical services. Performed routine system patching, security updates, and database maintenance to ensure compliance with security and performance SLAs.
- Collaborated with SRE, development, and QA teams to troubleshoot API integration issues, achieving a 20% improvement in system response times.
- Participated in Agile sprint planning and led the implementation of system enhancements with a focus on reliability and scalability.
- Authored internal documentation and facilitated knowledge transfer sessions, ensuring consistent understanding across engineering teams.
- Supported cloud migration efforts and hybrid cloud deployments, integrating cloud-native services (GCP) with legacy on-prem systems.
- Designed and integrated APIs to improve system interoperability and ensure seamless data exchange across distributed services.
- Engineered microservices-based architectures using Docker and Kubernetes, enhancing system modularity, fault tolerance, and horizontal scalability.
- Built and maintained CI/CD pipelines using Jenkins and GitLab, automating deployments and reducing software release cycles by 40%.
- Championed reliability-focused development practices by integrating observability tools and system health checks into deployment pipelines.

- **Research Assistant**
California State University, Chico

Feb 2022 – Feb 2023
Chico, California

- Spearheaded the development of cutting-edge neural network architectures using Python programming language, specialized libraries such as NumPy for numerical computing, TensorFlow for machine learning applications, and advanced techniques like transfer learning and recurrent neural networks.
- Implemented Kubernetes as a strategic tool for optimizing the deployment process of Word2Vec models, focusing on containerization, orchestration, and microservices architecture. This initiative involved fine-tuning deployment pipelines using Docker containers, Kubernetes clusters, and Helm charts to ensure seamless integration, scalability, high availability, and reliability.
- Adhered to industry best practices in DevOps and Agile methodologies, utilizing continuous integration/continuous deployment (CI/CD) pipelines, version control systems (e.g., Git), automated testing frameworks (e.g., pytest), and monitoring tools (e.g., Prometheus, Grafana) for efficient development, deployment, and maintenance of machine learning models.

- **Software Engineering Intern**
The Gund Company

Jun 2022 - Aug 2022
St Louis, Missouri

- Implemented custom PHP code to optimize websites and applications, ensuring compliance with technology governance guidelines.
- Managed content management systems (CMS) with a focus on security protocols and functionality standards for 70 days.
- Utilized WordPress CMS to analyze and resolve over 10 web-based system incidents, showcasing proficiency in client services and incident resolution methodologies. Implemented strategic solutions resulting in a 25% decrease in user-reported issues, improving system stability and user satisfaction.
- Led load balancing enhancements, increasing system throughput by 50%, reducing downtime by 30%, contributing to process improvement initiatives.

- **Tutor: Computer Science and Math**
California State University, Chico

Oct 2021 – May 2024
Chico, California

- Provided personalized tutoring to over 120 students in subjects spanning precalculus functions, mathematical graphs, analytical geometry, and advanced calculus, fostering deep understanding and academic success. Utilized a variety of teaching methodologies & technological tools to engage students.
- Conducted weekly one-on-one tutoring sessions for 30 students across various computer science courses, including Programming, Data Structures, and Advanced Algorithms and Complexity, enhancing their problem-solving abilities and conceptual grasp.
- Achieved a 100% satisfaction rate through detailed post-tutoring surveys, consistently receiving commendations for clear communication, expert subject knowledge, and fostering a supportive learning environment.

- Web Designing Intern** **April 2021 – June 2021**
Delhi, India
Sutraa Creations
 - Led the development of dynamic and responsive web designs, utilizing Docker for rapid prototyping and testing across diverse OS environments, ensuring seamless iteration and functionality. Utilized Angular for frontend development, resulting in improved website speed, reduced bounce rates, and a significant 30% increase in website traffic, enhancing user engagement metrics by 25%.
 - Implemented upgrades to key website elements using Angular, enhancing functionality and aesthetics, leading to a 15% increase in website speed, a 20% decrease in bounce rates, and a 35% improvement in user satisfaction based on feedback.
 - Independently resolved over 20 electronic framework troubleshooting issues within a 60-day period, showcasing strong problem-solving skills and autonomy. This effort led to a 40% reduction in system downtime, improved stability, and cost savings exceeding 10,000 INR.
- Web Development Intern** **Oct 2019 – Dec 2019**
Delhi, India
Unistall Systems Pvt. Ltd.
 - Collaborated with a cross-functional team of developers and GIS (Geographic Information System) professionals to design and develop two web applications, utilizing Agile methodologies, Git/GitHub for version control, and CI/CD pipelines, resulting in a significant 20% increase in productivity for over 500 employees. Integrated data from five services and databases, optimizing data retrieval processes with efficient SQL queries.
 - Contributed to GIS web applications development, using SSH for secure remote operations, Linux commands for server management, and pytest for automated testing, ensuring data integrity, system security, and high-quality code delivery.
 - Achieved a notable 25% reduction in data retrieval time. This initiative led to annual cost savings of \$1,000, enhanced operational efficiency, and demonstrated expertise in data integration, performance optimization, and cost-effective solutions.

PROJECTS

- Blue Connect - Mobile Application** **Aug 2023 – Dec 2023**
 - "Blue Connect" is designed as a mobile application with the primary goal of facilitating the connection of multiple devices via Bluetooth. This innovative app aims to streamline the process of device interconnectivity, enhancing user experience and enabling seamless data exchange across various platforms.
 - Spearheaded the conceptualization and development of an innovative mobile application, achieving a 20% increase in Bluetooth-based device connectivity efficiency. Led the creation of an innovative mobile app, boosting Bluetooth device connectivity efficiency by 20%.
 - Engineered crucial features such as seamless data exchange, multi-device connectivity, and an intuitive user interface, resulting in a 30% improvement in user experience ratings.
 - Leveraged advanced Bluetooth protocols, Android Studio, Xcode, Agile methodologies, UI/UX design principles, and mobile app testing frameworks to deliver a 25% reduction in bug occurrences and ensure a 95% user satisfaction rate.
 - <https://github.com/ChicoState/BlueConnect>
- Custom CNN Development** **Sep 2023 – Oct 2023**
 - Executed a sophisticated implementation of a Convolutional Neural Network (CNN) forward step entirely coded from scratch in C++, showcasing advanced proficiency in computational intelligence, deep learning, and software development.
 - Demonstrated a deep understanding of neural networks, deep learning frameworks, and their applications, highlighting expertise in machine learning algorithms and computer vision.
 - Developed complex algorithms utilizing CNN architecture, which is fundamental to modern machine learning and artificial intelligence systems.
 - Significance: Custom-built CNN represented a significant milestone, showcasing expertise in C++ programming, computational intelligence, neural network architectures, and algorithm optimization.
- Sentiment Analyzer: Movies** **October 2023 – December 2023**
 - Developed a comprehensive sentiment analysis solution using Python, focusing on movie review sentiment classification.
 - Leveraged Pandas, NumPy, and Scikit-learn for efficient data handling, numerical operations, and machine learning model implementation.
 - Applied Regular Expressions for data cleaning, TF-IDF vectorization for feature extraction, and Multinomial Naive Bayes for sentiment classification.
 - Visualized sentiment trends using Matplotlib, Seaborn, and WordCloud, providing actionable insights into review sentiments.
 - Outcome: Achieved accurate sentiment classification of movie reviews, showcasing the effectiveness of the Multinomial Naive Bayes model and the integration of advanced text processing and visualization techniques.
 - <https://github.com/JRobbinss18/Sentiment Analysis and Predict Python>
 - <https://docs.google.com/presentation/d/1GZDfMd40wUmSEuvBPkxkyELJHNUPIVfmSVV6MtkMdfk/edit#slide=id.p>